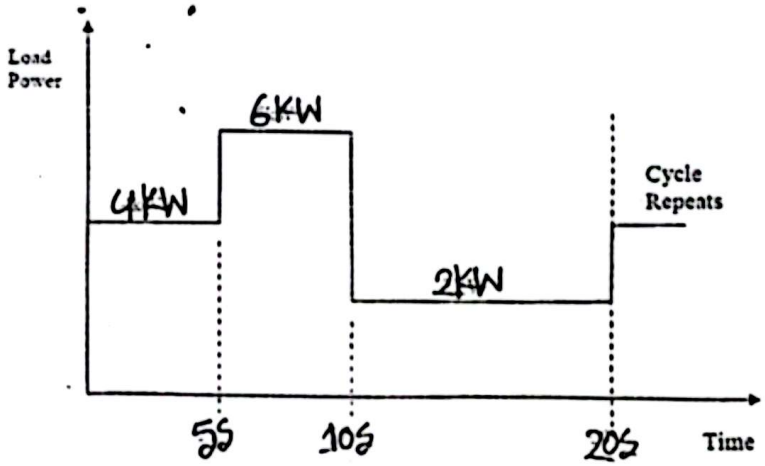
International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)
B.Sc. Engineering in CSE
Midterm Examination, Autumn 2025

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation
Time: 1.5 hours **Full Marks:** 30

Answer all the questions. The figures in the right-hand margin indicate full marks
Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

1 (a)	i. Discuss the Linear DC machine as a motor with a proper diagram. ii. A coil that has 700 turns develops an average induced voltage of 50 V. What must be the change in the magnetic flux that occurs to produce such a voltage if the time interval for this change is 0.7 seconds?	CO1	U	3+2
	OR			
1 (a)	i. Discuss the Linear DC machine as a generator with a proper diagram. ii. Discuss the Hysteresis loss and the Eddy current loss.	CO1	U	3+2
1 (b)	The following load torque profile was measured for a constant speed load. Choose an appropriately sized motor to drive this Load. Assuming a service factor of 30%. You do not need to consider starting performance. 	CO3	Ap	5

2 (a)	Classify the types of transformers. Analyze the open circuit test and short circuit test of the transformer with the necessary diagram.	CO3	An	5
2 (b)	A 50-kVA, 4,400/220-V transformer has $R_1 = 3.45\,\Omega$, $R_2 = 0.009\,\Omega$. The values of reactance are $X_1 = 5.2\,\Omega$ and $X_2 = 0.015\,\Omega$. Calculate for the transformer: (i) equivalent resistance as referred to both primary and secondary (ii) equivalent reactance as referred to both primary and secondary (iii) equivalent impedance as referred to both primary and secondary (iv) total Cu loss, using equivalent resistances as referred to each side.	CO3	An	5
	OR			
2 (b)	i. Why is transformer rating in VA or KVA? ii. A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results: O.C. Test (L.V. Side): 200 V, 1.2 A, 90 W S.C. Test (H.V. Side): 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO3	An	1+4
3 (a)	i. Describe the operation of the DC machine that converts electrical power into mechanical power with a proper net sketch. ii. State Faraday's laws of electromagnetic induction, Fleming's right-hand rule and Fleming's left-hand rule.	CO1	R	3+2
3 (b)	i. Draw the Equivalent Circuit and Simplified Equivalent Circuit of a DC Motor with proper notations. ii. Discover the several magnetic field characteristics of a permanent magnet and an electromagnet with net sketches.	CO1	Ap	2+3

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International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)

B.Sc. Engineering in CSE

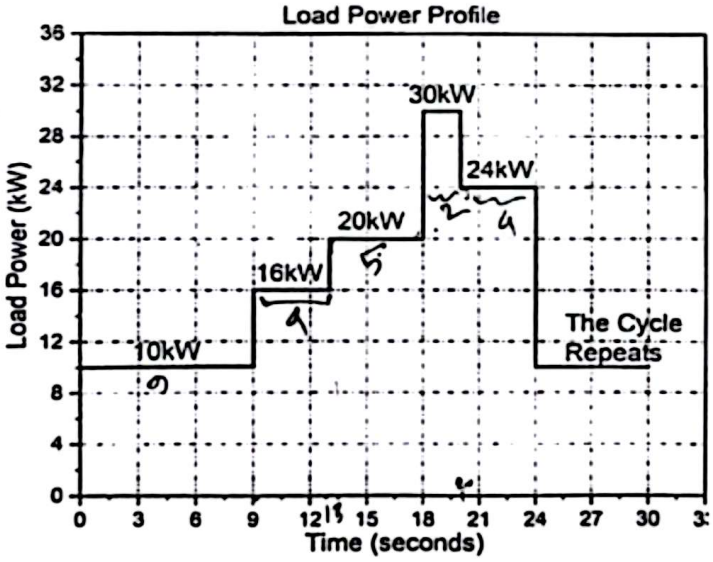
Midterm Examination, Spring 2025

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation

Time: 1.5 hours **Full Marks:** 30

Answer all the questions. The figures in the right-hand margin indicate full marks
Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

1(a)	i. List the different types of motor duty classes and explain any two of them with timing diagram.	CO1	Ap	5
	ii. The magnetic flux linked with a coil having 250 turns is changed from 1.4 Wb to 2 Wb in 0.45 seconds. Calculate the induced e.m.f in the coil.			
1(b)	The following load power profile was measured for a constant speed load. Choose an appropriately sized motor to drive this Load. Assuming a service factor of 30%. You do not need to consider starting performance.	CO3	Ap	5



2 (a)	i. What is transformer? Draw the equivalent circuit of transformer with proper notation.	CO1	R	5
	ii. Describe the core losses in a transformer.			

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2 (a)	i. Classify the transformer test	CO3	An	1+4
	ii. In no-load test of single-phase transformer, the following test data were obtained: Primary voltage: 230V; Secondary voltage: 115V Power Input: 40W; Primary Current: 0.5A Find the following: (i) the turns ratio (ii) the magnetizing component of no-load current (iii) its working component (iv) the iron loss			
	OR			
2 (b)	i. Distinguish between the transformer test. ii. A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results: O.C. Test (L.V. Side): 200 V, 1.2 A, 90 W S.C. Test (H.V. Side): 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO1	An	1+4
3 (a)	i. Describe the operation of DC machine that converts mechanical power into electrical power with proper net sketch. ii. State Faraday's laws of electromagnetic induction, Fleming's right-hand rule and Fleming's left-hand rule.	CO1	R	5
3 (b)	i. Why transformer test is needed? ii. A shunt generator delivers 455A at 230V and the resistance of the shunt field and armature are 40Ω and 0.03Ω respectively. Calculate the generated e.m.f.	CO3	An	1+4
	OR			
3 (b)	i. Analyze the concept of commutation. ii. A 255-V shunt motor runs at 1000 r.p.m at no-load and takes 9A. The total armature and shunt field resistances are respectively 0.2Ω and 255Ω. Calculate the speed when loaded and taking 50A. Assume the flux to be constant.	CO3	An	1+4

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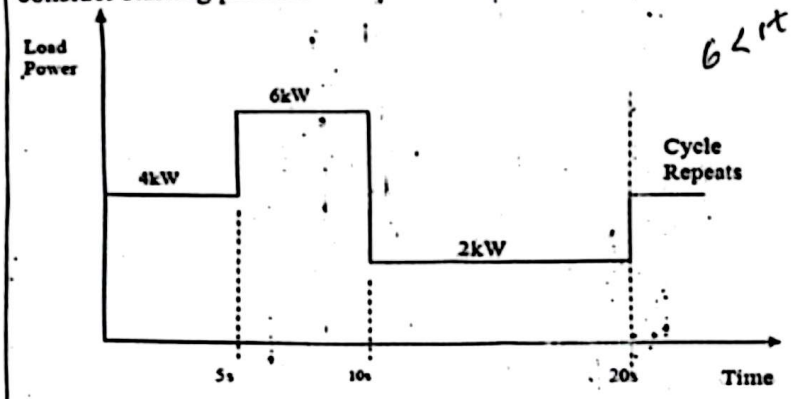
International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

Mid-term Examination, Spring-2024 | Program: BSc in CSE

Course Title: Electrical Drives and Instrumentation | Course Code: EEE-2421

Time: 1hour 30 minutes | Full Marks: 30

1(a)	Define Machine. Describe the operation of motor with proper net sketch.	CO1	R	4
1(b)	The load torque profile shown in figure 1(b) was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 25%. You do not need to consider starting performance.  Figure: 1(b)	CO2	An	3
1(c)	What is motor duty class? Write the different types of motor duty class with appropriate timing diagram.	CO1	U	3

2(a)	Define transformer. Why transformer is rated on VA?	CO1	R	2
2(b)	A single phase transformer has 400 primary and 1000 secondary turns. The net cross sectional area of the core is 60 cm^2. If the primary winding be connected to a 50 Hz supply at 520V, calculate- i) the peak value of flux density in the core ii) the voltage induced in the secondary winding.	CO1	An	3
2(c)	A 230/460 V transformer has a primary resistance of 0.2Ω and reactance of 0.5Ω and the corresponding values for the secondary are 0.75Ω and 1.8Ω respectively. Find the secondary terminal voltage when supplying 10A at 0.8 power factor (p.f.) lagging. **OR**	CO2	An	5

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	A 50kVA, 4400/220-V, 50-Hz transformer has a high voltage winding resistance (R1) of 3.45 Ω and a leakage resistance (X1) of 5.2Ω. The low voltage winding resistance (R2) is 0.009Ω and the leakage resistance (X2) is 0.015Ω. Find the equivalent winding resistance, reactance and impedance referred to the- i) high voltage side and ii) low voltage side.			

3(a)	Draw the equivalent circuit of transformer with proper notation.	CO1	R	2
3(b)	Write the faraday's law of electromagnetic induction. A coil that has 700 turns develops an average induced voltage of 50V. What must be the change in the magnetic flux occur to produce such a voltage if the time interval for this change is 0.7s.	CO1	An	3
3(c)	A shunt generator delivers 450 A at 230V and the resistance of the shunt field and armature are 50 Ω and 0.03 Ω respectively. Calculate the generated emf. **OR** *In no-load test of single phase transformer, the following test data were obtained: Primary voltage: 220V; Secondary voltage: 110V; Primary Current: 0.5A; Power Input: 30W. Resistance of the primary winding = 0.6 ohm* Find the following: i) The turn ratio ii) The magnetizing component of no load current iii) Its working (or loss) component iv) The iron loss.	CO2	An	5

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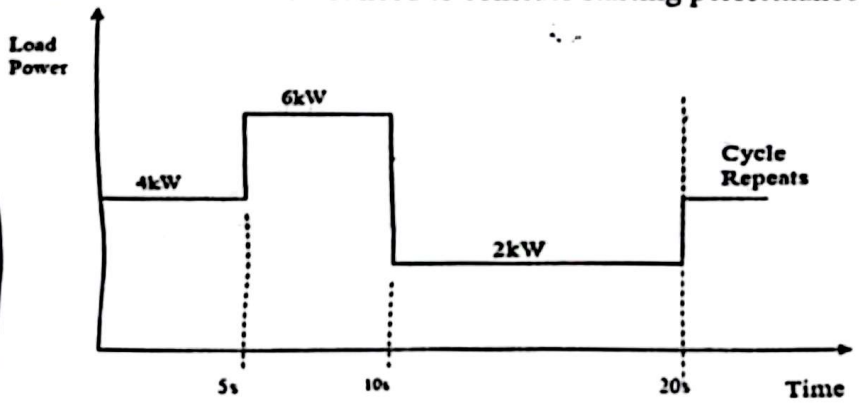
International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

Mid-term Examination, Autumn-2023 | Program: BSc in CSE

Course Title: Electrical Drives and Instrumentation | Course Code: EEE-2421

Time: 1hour 30 minutes | Full Marks: 30

1(a)	State the faraday's second law. Describe different types of motor duty class with appropriate timing diagram.	CO1	U	3
1(b)	What do you mean by motor sizing? The load torque profile shown in figure 1(b) was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 20%. You do not need to consider starting performance.  Figure: 1(b)	CO2	An	3
1(c)	Describe the operation of DC generator with proper net sketch.	CO1	R	4

2(a)	Draw the equivalent circuit of transformer with proper notation.	CO1	U	3
2(b)	When a transformer is connected to a 1000-V, 50-Hz supply the core loss is 1000 W, of which 650 is hysteresis and 350 is eddy current loss. If the applied voltage is raised to 2000 V and the frequency to 100 Hz, find the new core losses.	CO2	An	3
2(c)	In no-load test of single phase transformer, the following test data were obtained: Primary voltage: 220V; Secondary voltage: 110V; Primary Current: 0.5A; Power Input: 30W. Resistance of the primary winding = 0.6 ohm Find the following: i) The turn ratio ii) The magnetizing component of no load current iii) Its working (or loss) component iv) The iron loss.	CO2	An	4

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OR

2(a)	Define transformer. Discuss the working principle of a transformer.	CO1	R	3
2(b)	Why transformer test is needed? Compare the characteristics of an Ideal transformer and a practical transformer.	CO1	U	3
2(c)	A 5kVA 200/1000 V, 50Hz, single-phase transformer gave the following test result: O.C. Test (L.V. Side): 200V, 1.2A, 90W S.C. Test (H.V. Side): 50V, 5A, 110W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO2	An	4

3(a)	Describe the hysteresis losses and eddy current loss for transformer.	CO1	U	2
3(b)	Describe the working principle of linear DC machine as a Motor with appropriate figure and graph.	CO1	R	4
3(c)	A 30kVA, 2400/120-V, 50-Hz transformer has a high voltage winding resistance of 0.1 Ω and a leakage resistance of 0.22Ω. The low voltage winding resistance is 0.035Ω and the leakage resistance is 0.012Ω. Find the equivalent winding resistance, reactance and impedance referred to the i) high voltage side and ii) low voltage side	CO2	An	4

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International Islamic University Chittagong

Department of Computer Science and Engineering

B.S. in CSE

Mid term Exam, Spring 2023

Course Code: EEE-2421 Course Title: **Electrical Drives and Instrumentation**

Time: 90 minutes Full Marks: 30

(1) The figures in the right hand margin indicate full marks

(2) Course Outcomes and Bloom's Levels are mentioned in additional Columns

a	Analyze Linear DC Machine as a Motor.	CO1	E	4
	Explain hysteresis loss and eddy current loss in a ferromagnetic core?	CO1	U	3
	List the reasons for which it is important to size a motor? Discuss How to choose the right motor briefly?	CO1	Ap	3
	Compare the characteristics of an Ideal and a practical transformer. Briefly discuss the steps to eliminate the losses of a transformer?	CO2	U	3
	A 2200/200V transformer draws a no-load current of 0.6A and absorbs 400 watts. Find the magnetizing and iron loss currents.	CO2	An	3
	A 5 kVA 2000/1000 V, 50 Hz, single-phase transformer gave the following test results O.C. Test (L.V. Side) : 200 V, 1.2 A, 90 W S.C. Test (H.V. Side) : 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side	CO2	An	4
	Or			
	Discuss the working principle of a transformer and draw a vector diagram of an ideal transformer?	CO1	U	3
	A 50kVA, 4400/220 V transformer has $R_1 = 3.45\ \Omega$, $R_2 = 0.009\ \Omega$. The [illegible] resistances are $X_1 = 5.2\ \Omega$ and $X_2 = 0.015\ \Omega$ [illegible] transformer [illegible] as referred to primary [illegible] resistance referred to secondary [illegible] referred to both primary and secondary [illegible] referred to both primary and	CO2	An	4

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2	a)	In no-load test of a transformer, the following test data were obtained: Primary Voltage: 220V; Secondary voltage: 110V Power input: 30W; Primary Current: 0.5A Find the following: i) The turns ratio ii) Magnetising component of no-load current iii) Its working component iv) iron loss.	CO2	An	3
		Write a short note on the basic principle of a single loop DC generator.	CO1	U	2
3	b)	Briefly Discuss on armature reaction which occurs in electrical machines with necessary diagrams.	CO1	U	4
3	c)	A 250-V shunt motor runs at 1000 r.p.m. at no-load and takes 8A. The total armature and shunt field resistances are respectively 0.2 Ω and 250 Ω. Calculate the speed when loaded and take 50 A Assume the flux to be constant.	CO2	An	4

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